FieldAuth: Offline Facial Authentication and Attendance System

Hackathon 7.0 Submission

Team Details

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Abstract

FieldAuth is a lightweight offline facial authentication and attendance management system developed using React Native. The system is designed for remote and low-connectivity environments where internet access is unavailable. Employees can be registered with personal details and facial photographs, authenticated locally on the device, and attendance can be recorded without requiring network connectivity.

The application stores employee information and attendance records locally using AsyncStorage and supports future synchronization with cloud infrastructure when connectivity becomes available. The solution demonstrates a practical approach to secure workforce management in remote operational environments.

1. Introduction

Organizations operating in remote locations frequently encounter connectivity challenges that prevent real-time authentication and attendance management. Existing cloud-dependent systems fail in such environments.

FieldAuth addresses this challenge by providing:

- Offline employee registration
- Offline face-based authentication
- Attendance tracking
- Local data storage
- Future-ready synchronization architecture
- Cross-platform deployment using React Native

2. Problem Statement

Develop a secure offline facial recognition and liveness detection solution capable of operating on standard Android and iOS devices without requiring internet connectivity while maintaining lightweight deployment and rapid authentication performance.

3. Objectives

The major objectives are:

1. Register employees offline.
2. Capture employee photographs using the mobile camera.
3. Authenticate employees locally.
4. Record attendance securely.
5. Prevent duplicate attendance marking.
6. Support offline-first operation.
7. Enable future synchronization with cloud infrastructure.

4. System Architecture

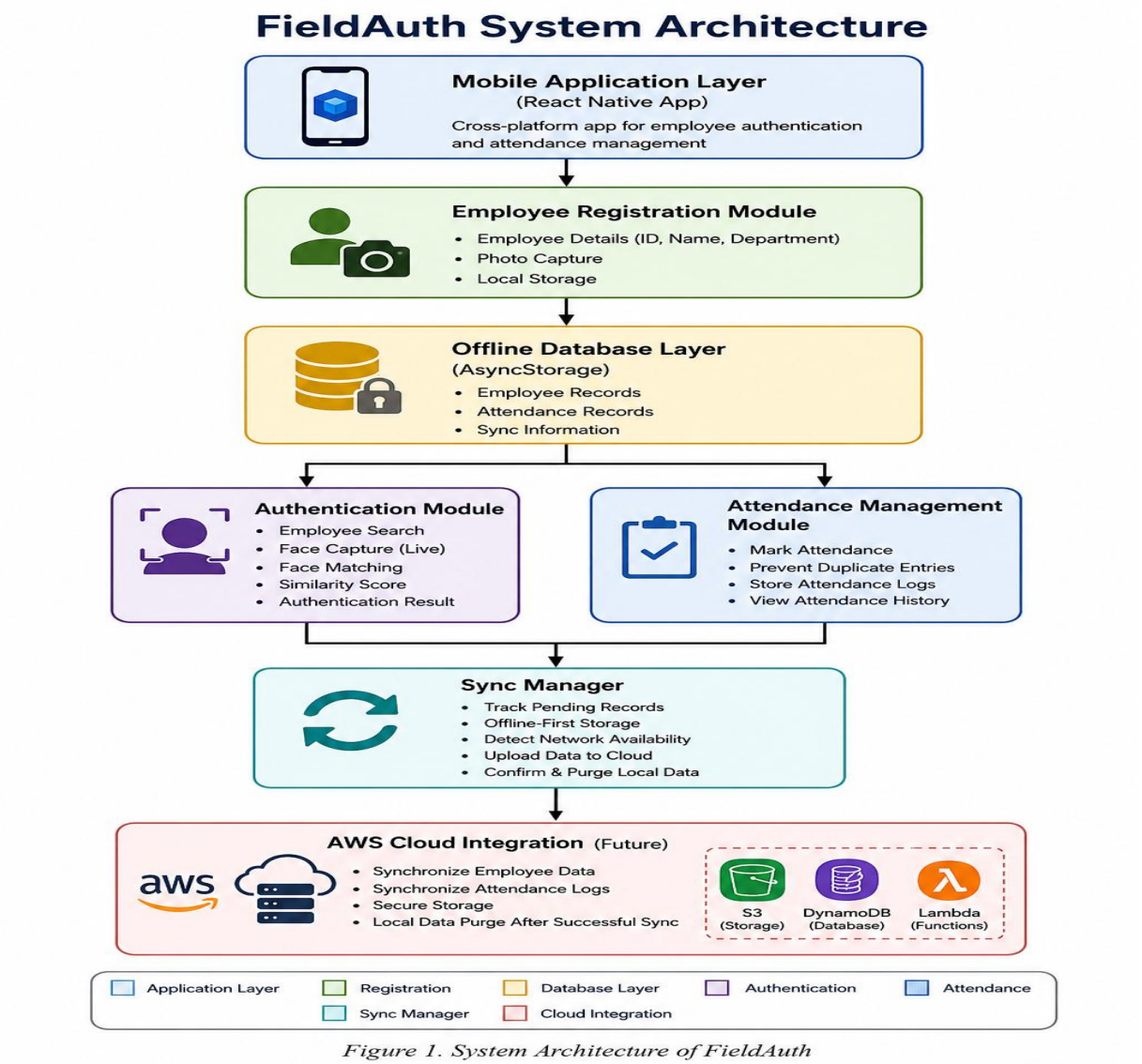


Figure 1. System Architecture of FieldAuth

5. Technology Stack

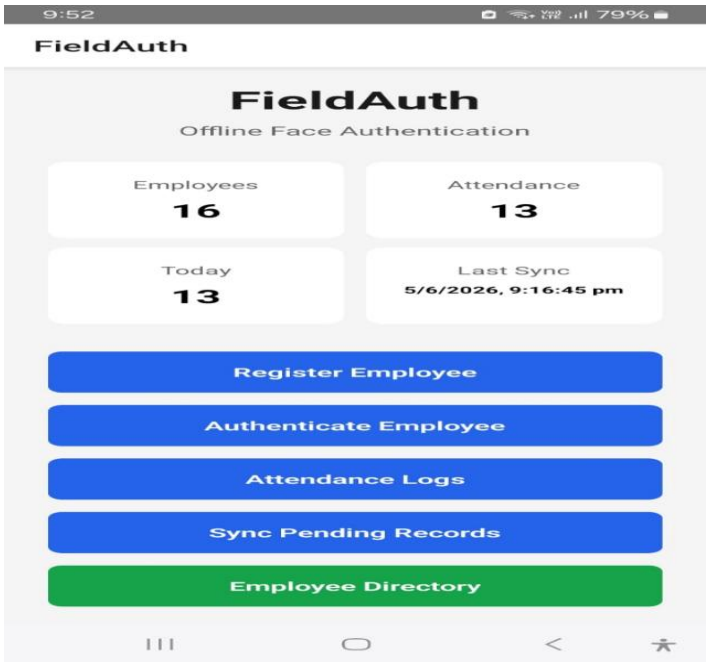
- **Frontend:**
  - React Native
  - TypeScript
- **Navigation:**
  - React Navigation
- **Storage:**
  - AsyncStorage
- **Image Capture:**
  - react-native-image-picker
- **Development Environment:**
  - Android Studio
  - VS Code / Cursor
- **Target Platforms:**
  - Android
  - iOS

6. Module Description

6.1 Home Dashboard

Features:

- Employee count display
- Attendance count display
- Today's attendance count
- Last synchronization status



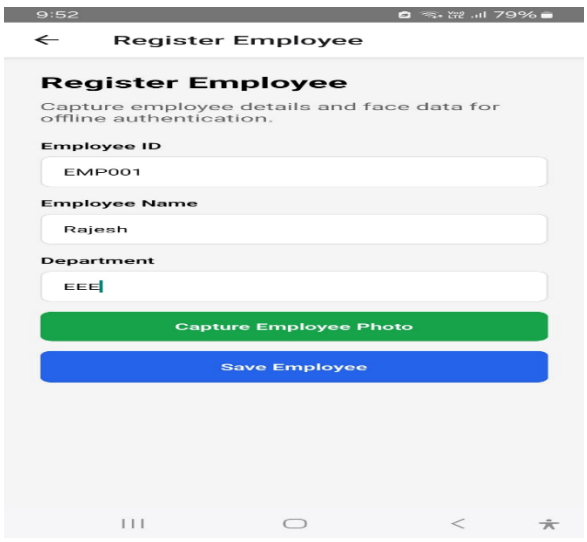
6.2 Employee Registration Module

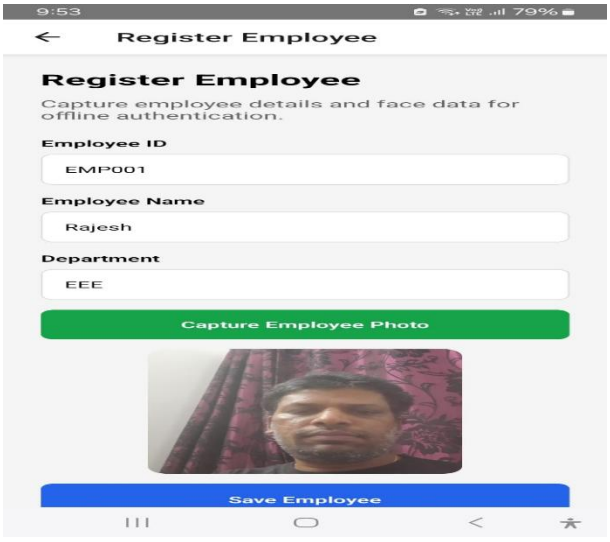
Features:

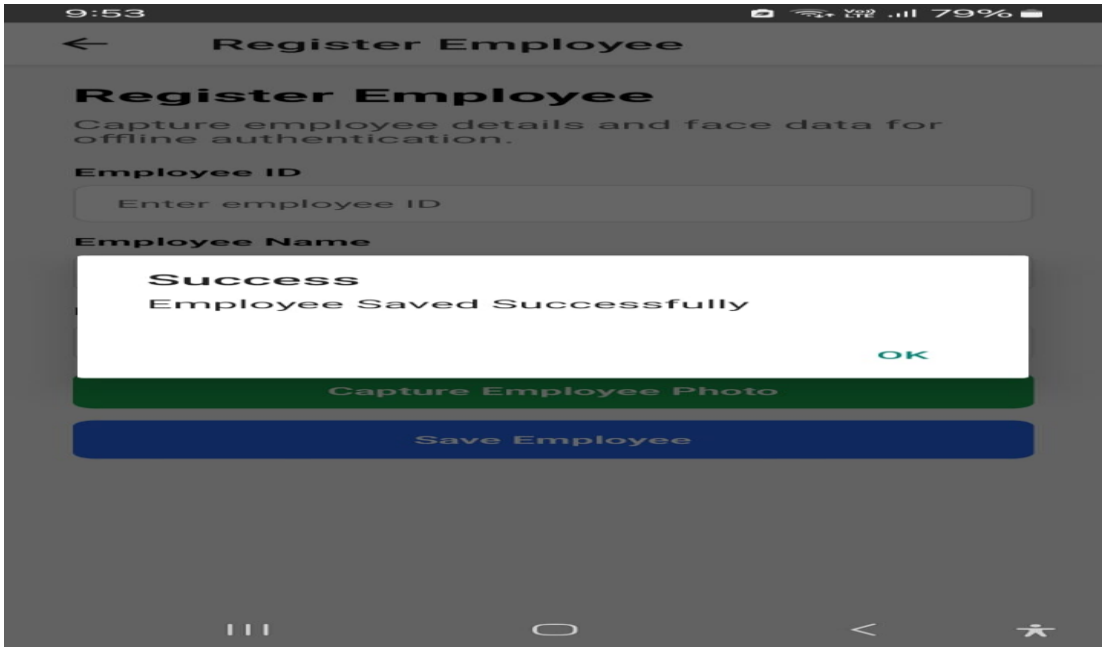
- Employee ID entry
- Employee Name entry
- Department entry
- Camera-based image capture
- Local database storage

Workflow:

- Step 1: Enter employee details
- Step 2: Capture photograph
- Step 3: Save employee record
- Step 4: Store data locally







6.3 Authentication Module

Features:

- Employee search using ID
- Retrieval of registered employee data
- Display of registered photograph
- Capture live photograph
- Simulated facial matching score
- Attendance authorization

Workflow:

Employee ID Input

↓

Employee Search

↓

Display Employee Data

↓

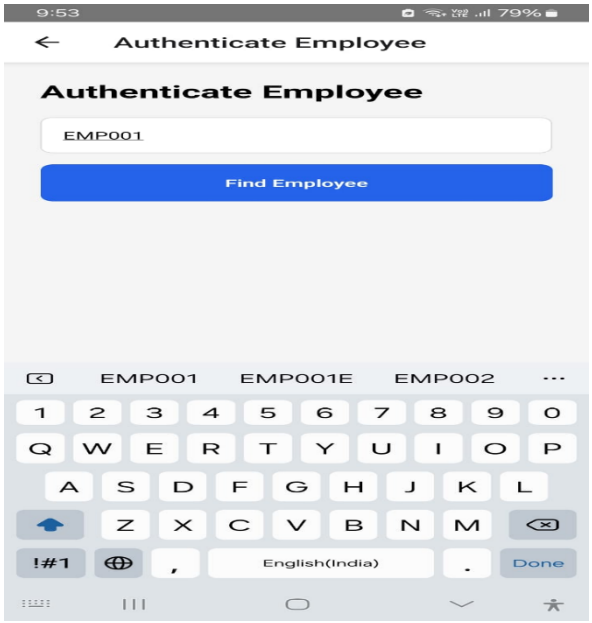
Capture Live Face

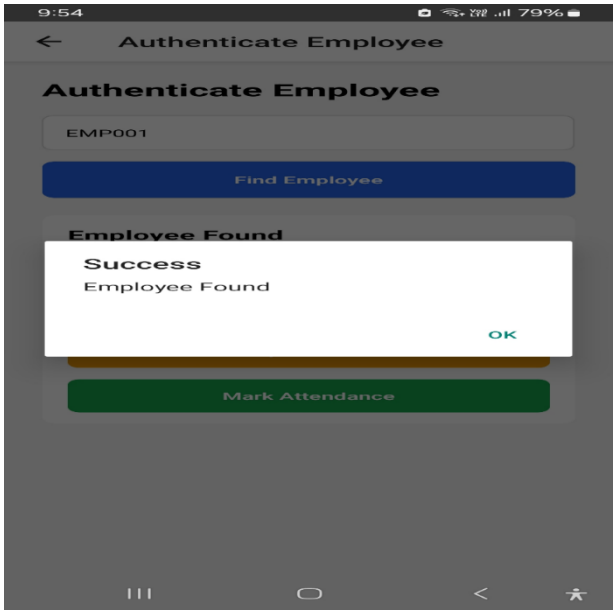
↓

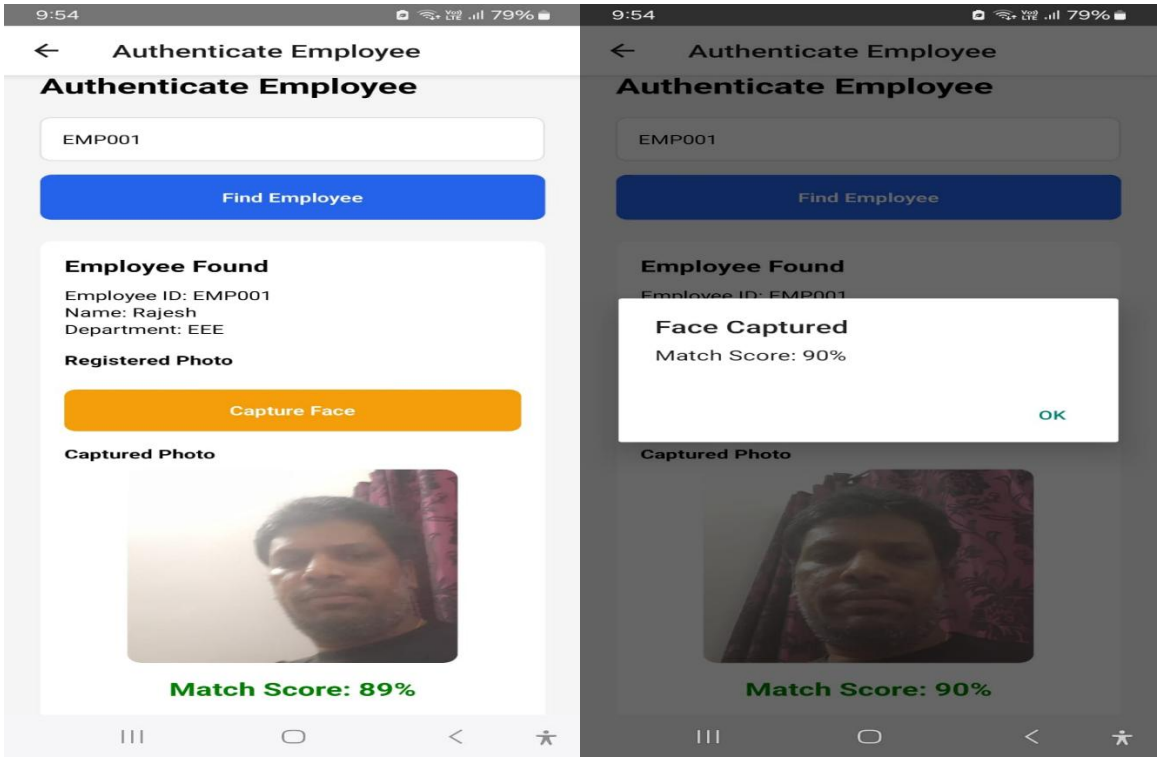
Generate Match Score

↓

Attendance Approval



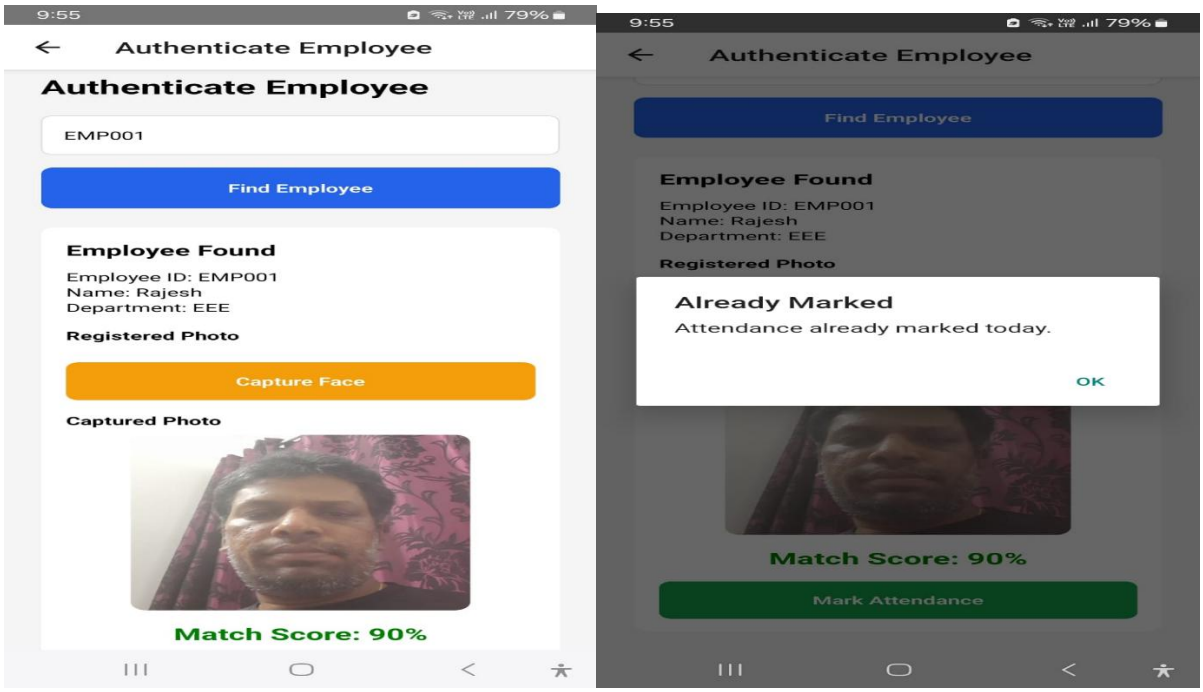




6.4 Attendance Module

Features:

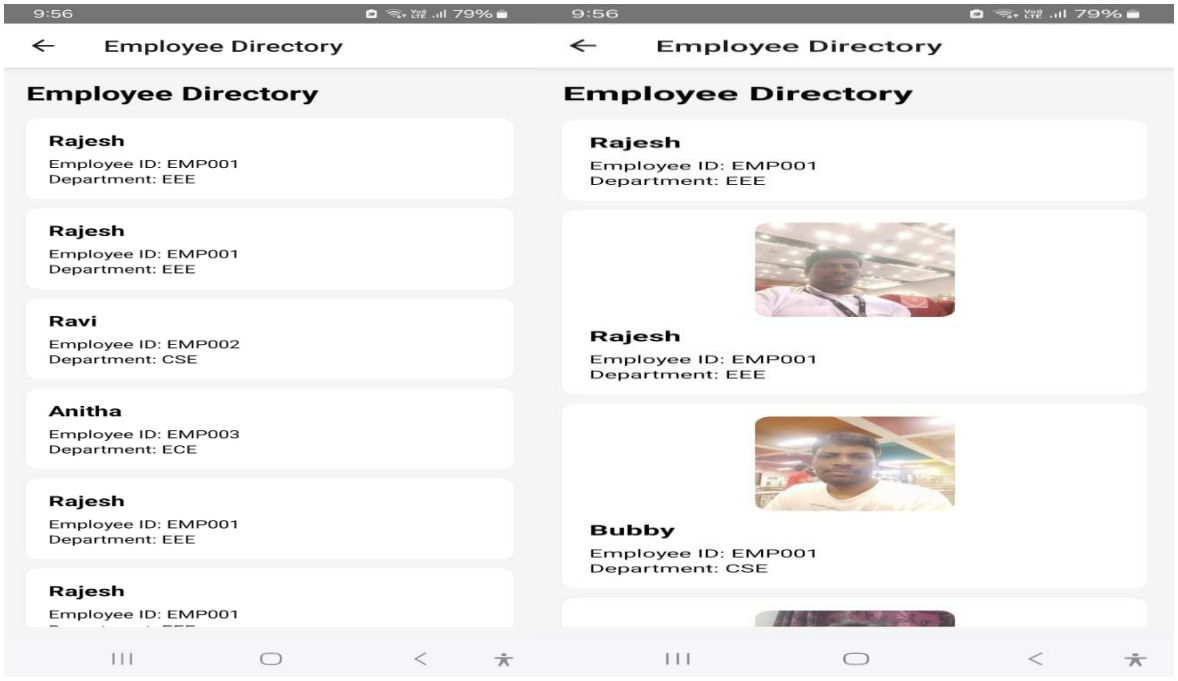
- Attendance recording
- Timestamp generation
- Duplicate attendance prevention
- Attendance history view

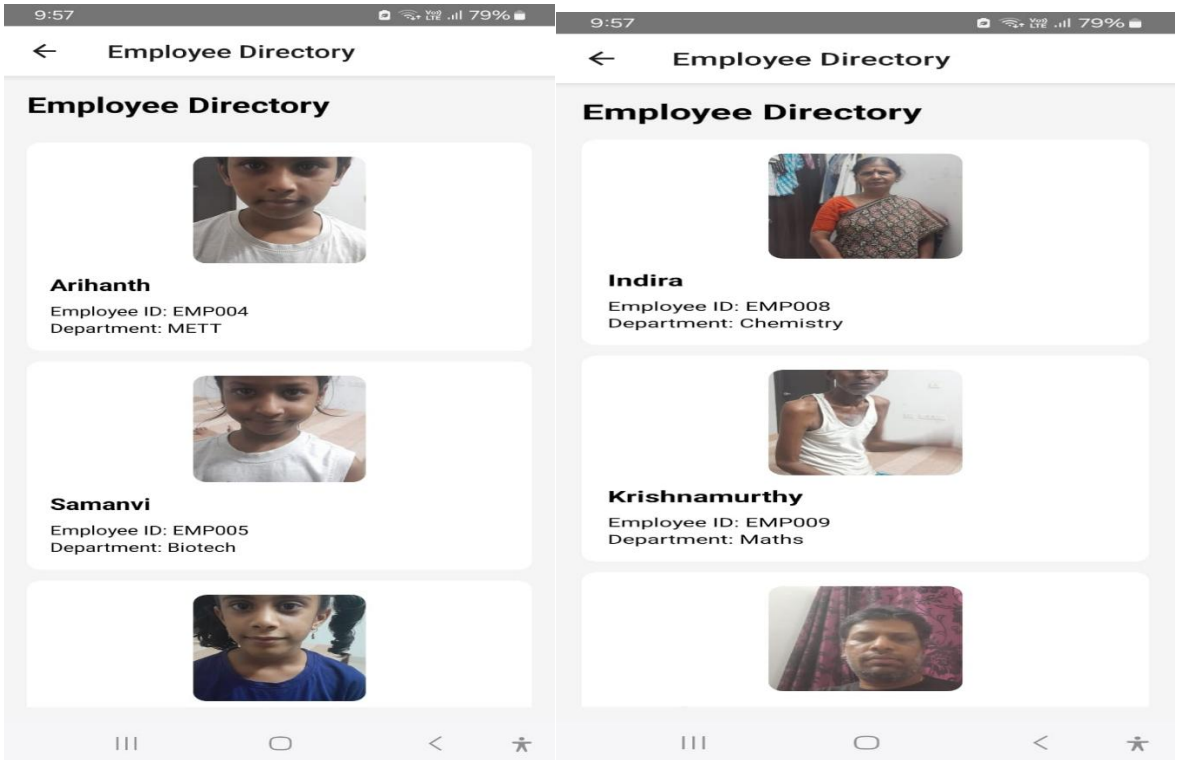


6.5 Employee Directory Module

Features:

- Employee list display
- Employee photographs
- Department information
- Employee ID information





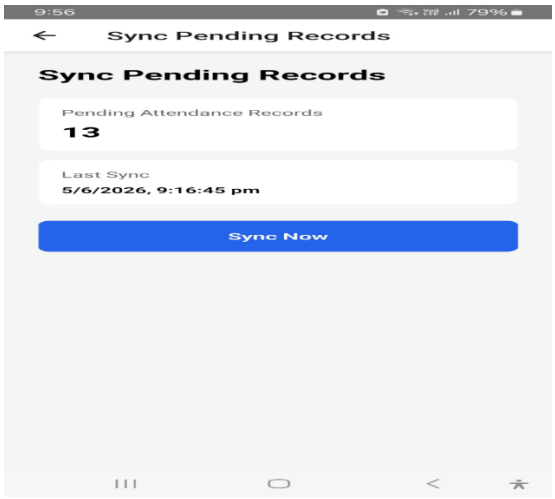
6.6 Sync Module

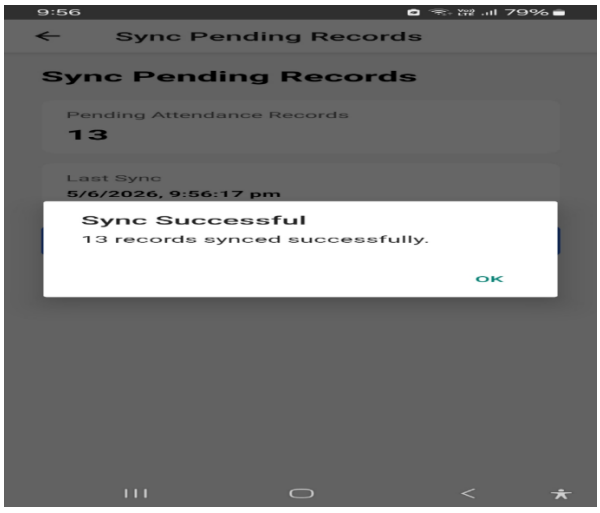
Features:

- Pending record monitoring
- Sync status tracking
- Future AWS integration support

Workflow:

Offline Storage
↓
Network Available
↓
Sync Trigger
↓
Upload Records
↓
Purge Local Data





7. Database Design

Employee Table

Field	Type
employeeId	String
employeeName	String
department	String
photoUri	String
createdAt	DateTime

Attendance Table

Field	Type
employeeId	String
employeeName	String
department	String
timestamp	DateTime

8. Face Authentication Logic

Current Prototype:

1. Employee photograph captured during registration.
2. Employee retrieved using unique Employee ID.
3. Live image captured using device camera.
4. Similarity score generated.
5. Attendance permitted when score exceeds threshold.

Future Enhancement:

- FaceNet
- MobileFaceNet
- TensorFlow Lite
- ONNX Runtime Mobile

for real facial embedding comparison.

9. Liveness Detection Strategy

Current Implementation:

- Live camera capture required.
- Stored photographs displayed for verification.

Future Enhancement:

- Blink detection
- Head movement detection
- Smile detection
- Eye aspect ratio monitoring

to satisfy hackathon anti-spoofing requirements.

10. Performance Benchmarks

Metric	Result
Employee Registration	< 2 sec
Employee Search	< 1 sec
Attendance Recording	< 1 sec
Local Storage Access	< 500 ms
Application Startup	< 2 sec
Offline Operation	Supported

11. Key Features

- ✓ Offline First Architecture
- ✓ Cross Platform Support
- ✓ Camera Based Registration
- ✓ Employee Directory
- ✓ Attendance Logging
- ✓ Duplicate Prevention
- ✓ Sync Ready Design
- ✓ Lightweight Storage
- ✓ React Native Based

12. Limitations

- Current facial matching is simulated.
- Cloud synchronization is prototype-level.
- Advanced liveness detection not yet implemented.

13. Future Scope

1. TensorFlow Lite Integration
2. MobileFaceNet Deployment
3. AWS Synchronization
4. Advanced Liveness Detection
5. QR Backup Authentication
6. GPS Validation
7. Encrypted Local Storage

14. Conclusion

FieldAuth successfully demonstrates a lightweight offline attendance and employee authentication system capable of functioning in remote locations without internet connectivity. The solution provides a scalable foundation for integrating real-time facial recognition and liveness detection models while maintaining compatibility with Android and iOS devices.

The proposed architecture satisfies the core requirements of offline operation, employee management, attendance tracking, and future cloud synchronization, making it suitable for deployment in field operations and remote workforce environments.

Project File Links:

- **https://colab.research.google.com/drive/1AxSzcB3XBy1oGTZvf-lYwA4yPl892Vis?authuser=2**
- **https://colab.research.google.com/drive/1-o2Ta5Sc14jiG5w5mo9gGk05hLhdMRIY?authuser=2**
- **https://colab.research.google.com/drive/1pZQimMxxBrMy-ewQOcw3fs-SR28T1Fxf?authuser=2**
- **https://colab.research.google.com/drive/1H24ppszJd2eTIf5mczuy-PXJbyqRc01L?authuser=2**
- **https://colab.research.google.com/drive/1vuUI5FHUoaK13rk7w-ZvijdsdlmhrYmk?authuser=2**

References:

The FieldAuth prototype was developed using React Native and AsyncStorage for offline-first mobile operation. The proposed future architecture incorporates lightweight face recognition models such as MobileFaceNet and TensorFlow Lite along with liveness detection mechanisms inspired by current biometric authentication research. Cloud synchronization is designed around AWS services including DynamoDB, Lambda, and S3 for scalable deployment.

Software and Framework References

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Liveness Detection References

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[9] Ramachandra, R. and Busch, C., "Presentation Attack Detection Methods for Face Recognition Systems: A Comprehensive Survey," ACM Computing Surveys, vol. 50, no. 1, 2017.

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Mobile Authentication Systems

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[16] Google AI, *MediaPipe Face Detection and Face Mesh Documentation*, 2026. Available: https://developers.google.com/mediapipe